

KENOSHA AP, WI, USA

WMO: 726505

Lat: 42.595N

Lon: 87.938W

Elev: 743

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04845

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) 1.3	(d) -12.4	(e) 2.9	(f) -2.4	(g) -8.0	(h) 3.7	(i) 2.7	(j) 27.0	(k) 25.8	(l) 24.8	(m) 26.5	(n) 9.7	(o) 280	(p) 0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	90.1	74.5	87.1	73.1	84.0	71.4	77.1	86.9	75.2	83.7	73.4	81.2	12.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	128.9	82.2	72.4	123.3	80.5	70.9	117.1	78.7	40.9	86.9	39.1	83.9	37.5	81.5	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.8	(b) 21.4	(c) 19.1	(e) -10.7	(f) 95.1	(g) 6.1	(h) 4.2	(i) -15.1	(j) 98.1	(k) -18.7	(l) 100.6	(m) -22.1	(n) 102.9	(o) -26.6	(p) 105.9		
(a) 24.8	(b) 21.4	(c) 19.1	(e) -11.3	(f) 79.9	(g) 6.0	(h) 2.2	(i) -15.6	(j) 81.5	(k) -19.1	(l) 82.8	(m) -22.4	(n) 84.1	(o) -26.8	(p) 85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	22.7	25.8	35.4	46.4	57.1	66.6	71.8	70.2	63.6	51.7	39.5	28.7
	DBStd	19.33	11.69	11.25	10.99	8.79	8.76	7.47	6.31	5.36	7.80	8.78	9.60	10.78
	HDD50	3282	845	680	473	170	28	1	0	0	2	87	334	662
	HDD65	6693	1310	1098	919	563	278	68	12	13	119	423	765	1125
	CDD50	2705	0	2	21	62	248	500	676	626	410	138	20	2
	CDD65	639	0	0	1	4	33	118	223	174	77	9	0	0
	CDH74	5744	0	0	11	67	341	1110	2126	1379	638	72	0	0
	CDH80	1773	0	0	1	12	82	351	743	393	181	10	0	0
Wind	WSAvg	9.3	10.7	10.5	10.3	11.0	9.6	8.1	7.4	6.9	7.6	9.2	10.0	10.2
Precipitation	PrecAvg	34.6	1.7	1.5	2.2	3.7	3.8	4.1	3.3	3.7	3.2	2.8	2.6	1.9
	PrecMax	41.6	3.7	3.5	4.4	7.1	10.6	8.5	7.3	11.2	6.9	6.5	5.4	4.1
	PrecMin	24.0	0.4	0.2	0.6	1.8	0.6	0.7	1.7	1.4	0.7	0.9	0.6	0.6
	PrecStd	4.6	0.9	0.9	1.2	1.5	2.1	2.0	1.3	1.9	1.5	1.5	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	60.8	73.5	81.4	86.8	92.3	96.6	92.4	89.6	81.3	68.5	59.5
		MCWB	49.5	52.9	62.1	63.0	69.6	74.1	77.3	75.5	73.0	68.6	58.4	55.3
	2%	DB	46.5	51.7	64.4	73.5	82.2	88.0	90.5	87.9	85.0	75.0	63.1	53.5
		MCWB	41.5	45.8	56.1	59.5	67.2	72.4	76.0	74.1	71.4	64.2	54.3	49.7
	5%	DB	40.9	44.9	58.5	68.0	78.0	84.0	87.5	84.1	81.3	70.6	58.6	46.8
		MCWB	37.6	40.4	51.3	56.5	65.5	71.3	73.9	72.1	68.6	61.7	52.1	43.3
	10%	DB	37.4	39.5	51.8	62.6	73.1	80.8	83.9	81.3	77.2	66.0	54.3	43.1
		MCWB	35.2	36.1	46.0	52.5	62.6	69.3	72.0	70.4	67.3	58.6	48.5	39.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	54.2	63.0	66.0	73.0	77.4	80.5	77.9	76.3	71.7	60.3	55.4
		MCDB	53.1	57.0	73.1	78.2	82.9	87.6	90.2	88.7	85.1	78.6	65.6	58.1
	2%	WB	42.0	46.0	57.7	61.3	69.5	74.8	77.9	75.8	73.3	66.6	56.7	50.0
		MCDB	45.7	49.9	63.5	71.4	79.1	84.4	87.8	84.4	80.8	71.9	61.8	52.3
	5%	WB	38.1	40.5	51.9	57.7	67.1	72.7	75.8	74.0	71.2	63.2	52.7	44.1
		MCDB	40.6	44.7	56.9	66.2	75.2	81.7	84.7	81.5	78.2	68.2	57.1	46.7
	10%	WB	35.4	36.8	45.8	53.0	64.3	70.7	73.9	72.3	68.9	59.8	48.7	39.6
		MCDB	37.6	39.5	51.2	61.2	71.8	78.6	81.7	78.8	75.4	65.5	54.1	42.6
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.8	16.1	18.8	19.9	19.7	20.0	19.3	20.3	18.0	15.4	13.1
		MCDBR	16.5	19.4	24.2	28.6	25.5	23.0	22.0	22.5	23.7	23.8	21.2	16.7
		MCWBR	13.6	14.7	17.3	16.9	14.4	11.7	10.2	10.7	11.9	14.2	15.1	13.9
	5% WB	MCDBR	15.3	18.8	21.9	26.0	22.8	20.6	19.6	18.9	20.6	20.1	19.0	15.2
MCWBR		13.2	15.0	17.1	17.4	14.4	12.0	10.7	10.3	11.5	14.6	15.4	13.9	
Clear-Sky Solar Irradiance	taub	0.290	0.309	0.341	0.392	0.416	0.435	0.433	0.425	0.395	0.348	0.310	0.289	
	taud	2.430	2.382	2.393	2.279	2.262	2.257	2.263	2.284	2.341	2.472	2.519	2.474	
	Ebn at Noon	271	283	286	278	273	267	266	265	264	266	263	262	
	Edh at Noon	24	30	33	40	42	43	42	40	35	27	22	21	
All-Sky Solar Radiation	RadAvg	474	760	1130	1458	1679	1919	1946	1684	1369	884	553	390	
	RadStd	63	90	98	121	130	128	90	104	116	101	54	42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page